

PRACTICAL_WEEK-02

TASK 1:

1. pwd

```
Enter a Linux command: pwd
```

```
Parent Process
```

```
Parent PID : 6025
```

```
Child PID : 6028
```

```
Child Process
```

```
Child PID : 6028
```

```
Parent PID : 6025
```

```
Executing command: pwd
```

```
/Users/amrutha/OSSP
```

```
Child process completed.
```

2.Uname

```
Enter a Linux command: uname
```

```
Parent Process
```

```
Parent PID : 6217
```

```
Child PID : 6219
```

```
Child Process
```

```
Child PID : 6219
```

```
Parent PID : 6217
```

```
Executing command: uname
```

```
Darwin
```

```
Child process completed.
```

3.Ls

```
Enter a Linux command: ls
```

```
Parent Process
```

```
Parent PID : 5885
```

```
Child PID : 5901
```

```
Child Process
```

```
Child PID : 5901
```

```
Parent PID : 5885
```

```
Executing command: ls
```

```
assets      docs      LICENSE    obj        scripts    tests
bin          include  Makefile   README.md  src
```

```
Child process completed.
```

```
amrutha@Amruthas-MacBook-Air OSSP %
```

4. Whoami

```
Enter a Linux command: whoami
```

```
Parent Process
```

```
Parent PID : 6158
```

```
Child PID : 6168
```

```
Child Process
```

```
Child PID : 6168
```

```
Parent PID : 6158
```

```
Executing command: whoami
```

```
amrutha
```

```
Child process completed.
```

5. Date

```
Enter a Linux command: date
```

```
Parent Process
```

```
Parent PID : 6085
```

```
Child PID : 6086
```

```
Child Process
```

```
Child PID : 6086
```

```
Parent PID : 6085
```

```
Executing command: date
```

```
Mon Aug 3 10:38:38 IST 2026
```

```
Child process completed.
```

TASK 2:

1.uname

```
amrutha@Amruthas-MacBook-Air OSSP % uname -a
Darwin Amruthas-MacBook-Air.local 25.5.0 Darwin Kernel Version 25.5.0: Tue Jun  9 22:26:22 PDT 2026; root:xnu-12377.121.10~1/RELEASE_ARM64_T8132 arm64
```

2.ps

```
[amrutha@Amruthas-MacBook-Air OSSP % ps
  PID TTY          TIME CMD
 5621 ttys003      0:00.06 -zsh
```

3.lscpu

```
amrutha@Amruthas-MacBook-Air OSSP % system_profiler SPHardwareDataType
2026-08-03 11:01:30.593 system_profiler[9897:5189053] hw.cpubfamily: 0x6f5129ac
Hardware:

  Hardware Overview:

    Model Name: MacBook Air
    Model Identifier: Mac16,12
    Model Number: MW0X3LL/A
    Chip: Apple M4
    Total Number of Cores: 10 (4 Performance and 6 Efficiency)
    Memory: 16 GB
    System Firmware Version: 18000.121.3
    OS Loader Version: 18000.121.3
    Serial Number (system): KH6QD77CJF
    Hardware UUID: 8D84ABB8-B0EA-57A8-8836-DBD978C83FF7
    Provisioning UDID: 00008132-001C505E0C8B801C
    Activation Lock Status: Disabled
```


All appear to run simultaneously because the OS rapidly switches the CPU between them.

2. Memory Abstraction

The operating system manages RAM through **virtual memory**.

Functions include:

- Allocating memory to processes
- Protecting one process from accessing another's memory
- Reclaiming unused memory
- Using swap space when RAM becomes full

Applications only see their allocated memory rather than the physical RAM.

3. Storage Abstraction

Storage devices such as SSDs and HDDs are managed through a **file system**.

The operating system:

- Creates files and folders
- Organizes data
- Reads and writes files
- Controls access permissions

Users interact with files instead of physical disk sectors.

4. I/O Device Abstraction

Input/output devices include:

- Keyboard
- Mouse
- Printer
- USB devices
- Monitor
- Network card

The operating system communicates with these devices through **device drivers**.

Applications simply request services like:

- Read keyboard input

- Display text
- Print a document

The OS handles all hardware communication internally.

Role of the Operating System

The operating system acts as an interface between the user and hardware by:

- Managing CPU scheduling
- Managing memory allocation
- Organizing files and storage
- Controlling input/output devices
- Providing security and protection
- Managing multiple processes simultaneously

Observation

Using Linux commands, we observed that:

- `uname` provides kernel and operating system information.
- `lscpu` displays processor details.
- `ps` lists active processes managed by the OS.
- `top` shows real-time CPU and memory usage.

These commands demonstrate how the operating system manages hardware resources efficiently while allowing multiple applications to run smoothly.

Conclusion

The Linux operating system abstracts hardware resources by providing high-level services for CPU scheduling, memory management, storage organization, and device communication. This abstraction hides hardware complexity from users and applications, making the system easier to use, secure, and efficient. The commands `uname`, `lscpu`, `ps`, and `top` help observe how the operating system interacts with and manages the underlying hardware.